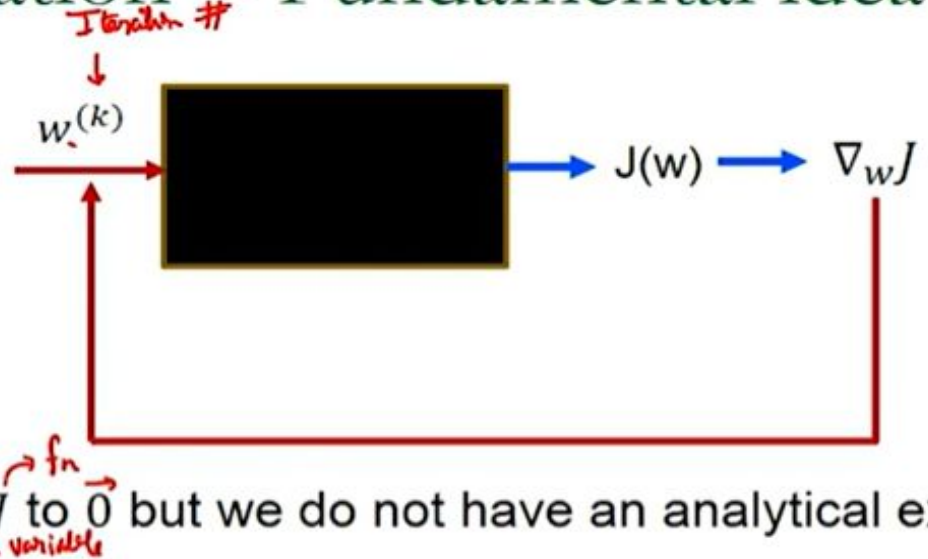


NUMERICAL GRADIENT

Iterative optimization -- Fundamental idea

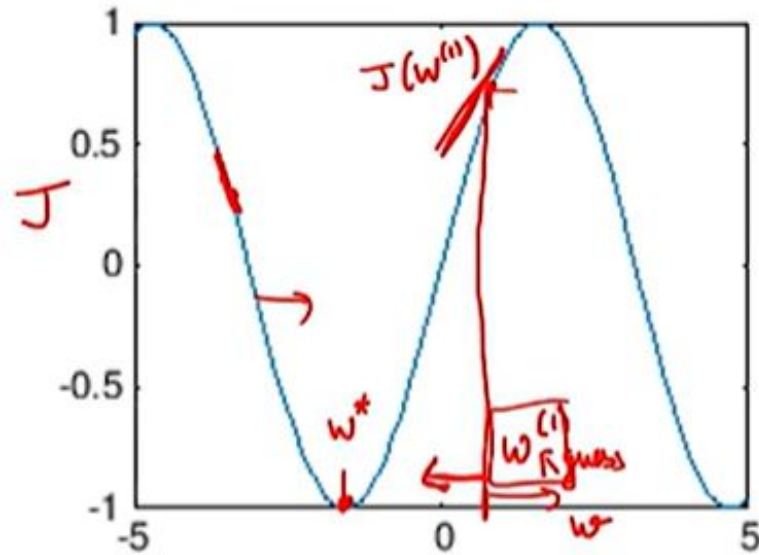


- We want to drive $\nabla_w J$ to 0 but we do not have an analytical expression.

Iterative Process

- Guess for w
- Run through the black box and find **value** of $J(w)$
 - This value may be obtained through a program instead of an expression
- Find $\nabla_w J$
 - We will discuss methods for determining $\nabla_w J$ numerically in later videos
- If $\nabla_w J = 0$, we stop, else we need to take a new guess
 - More precisely, improve our guess
- A very common method for improving guess is called **Gradient Descent**

Gradient Descent (Scalar case)

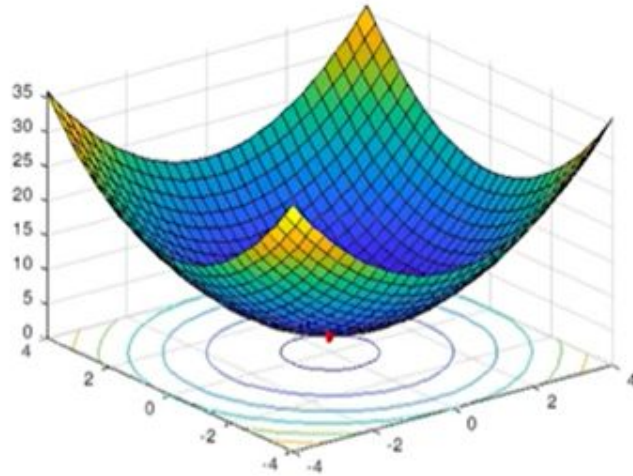


Scalar.
 $J(w)$
 $\frac{dJ}{dw} \neq 0$
 $\frac{dJ}{dw} > 0$
 $w = w - \alpha \left(\frac{dJ}{dw} \right)$

- Our task is to improve our guess for w such that we move from a region of higher gradient to a region of lower gradient
- For scalar (i.e. one component) w , this is easy

$$w^{new} = w^{old} - \alpha \left(\frac{dJ}{dw} \right)$$

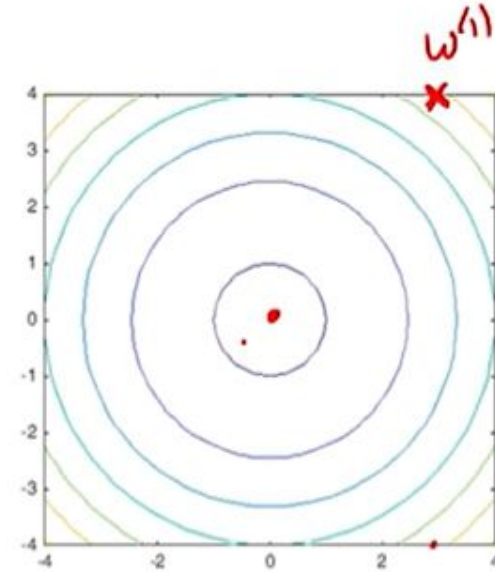
Gradient Descent example



$$J(w) = w_1^2 + w_2^2 + 4$$

$$\nabla_w J(w) = \begin{bmatrix} 2w_1 \\ 2w_2 \end{bmatrix}$$

$$\vec{w}^{new} = \vec{w}^{old} - \alpha \nabla_w J$$
$$w_1^{new} = w_1^{old} - \alpha (2w_1)$$



Gradient Descent gives the iterative formula

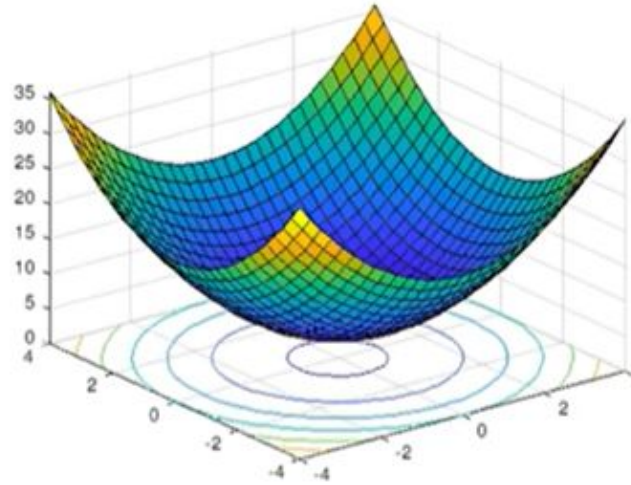
$$w_1^{(k+1)} = w_1^{(k)} - \alpha (2w_1^{(k)})$$

$$w_2^{(k+1)} = w_2^{(k)} - \alpha (2w_2^{(k)})$$

We know that the actual minimum is at $w = [0 \ 0]^T$

Let us start with an initial guess of $w^0 = [3 \ 4]^T$

Gradient Descent example

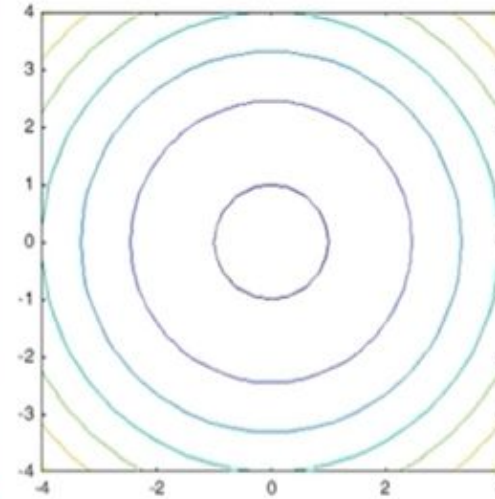


$$J(w) = w_1^2 + w_2^2 + 4$$

$$\nabla_w J(w) = \begin{bmatrix} 2w_1 \\ 2w_2 \end{bmatrix}$$

$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

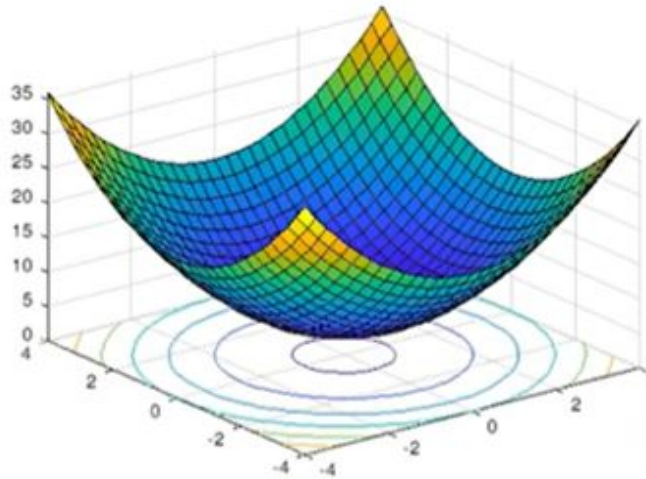
$$w_2^{k+1} = w_2^k - \alpha (2w_2^k)$$



$$\mathbf{w}^0 = [3 \ 4]^T \quad \alpha = 2$$

Iteration (k)	$\mathbf{w}^k$	$\nabla_w J = 2[\mathbf{w}_1 \ \mathbf{w}_2]$	J	$\mathbf{w}^{k+1} = \mathbf{w}^k - \alpha \nabla_w J$
0 <i>Ini</i>				

Gradient Descent example

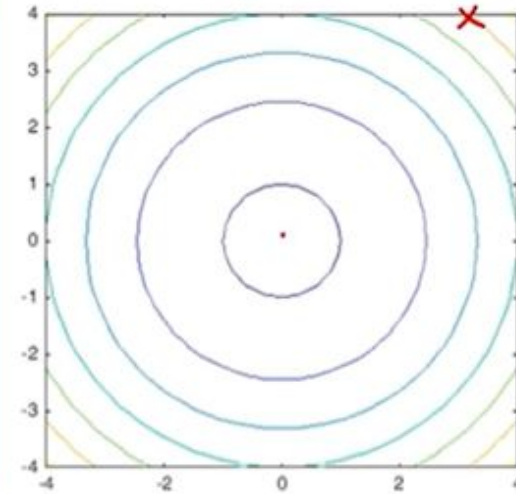


$$J(w) = w_1^2 + w_2^2 + 4$$

$$\nabla_w J(w) = \begin{bmatrix} 2w_1 \\ 2w_2 \end{bmatrix}$$

$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

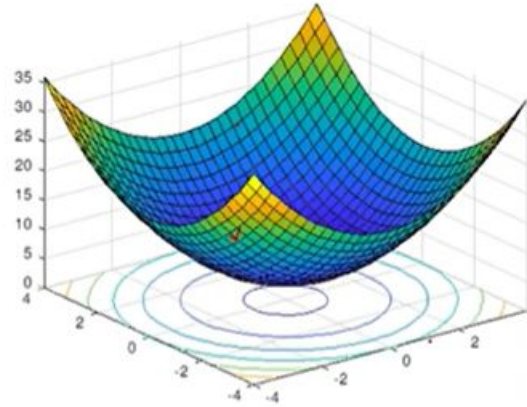
$$w_2^{k+1} = w_2^k - \alpha (2w_2^k)$$



$$w^0 = [3 \ 4]^T \quad \underline{\alpha = 2}$$

Iteration (k)	w^k	$\nabla_w J = 2[w_1 \ w_2]$	J	$w^{k+1} = w^k - \alpha \nabla_w J$
0 <i>Init guess</i>	[3 4]	[6 8]	29	$[3 \ 4] - 2 * [6 \ 8] = [-9 \ -12]$

Gradient Descent example

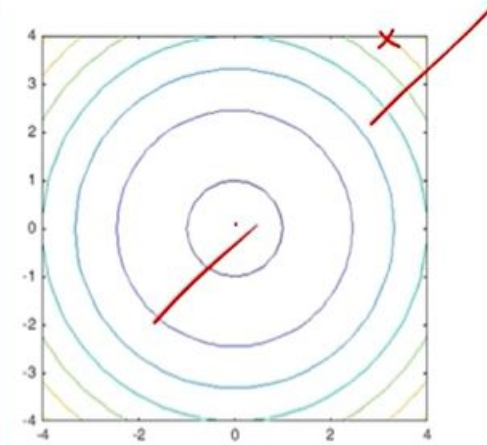


$$J(w) = w_1^2 + w_2^2 + 4$$

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$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

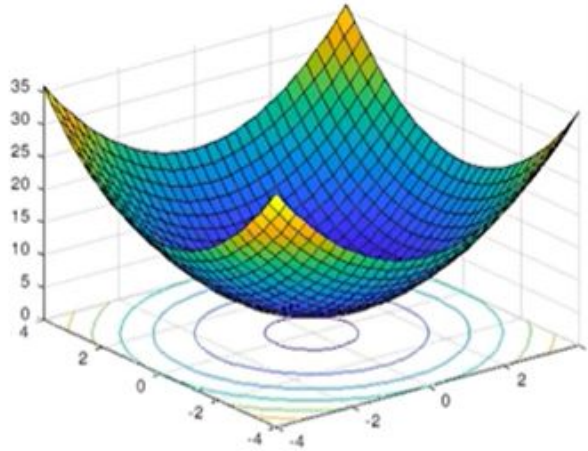
$$w_2^{k+1} = w_2^k - \alpha (2w_2^k)$$



$$w^0 = [3 \ 4]^T \quad \underline{\alpha = 2}$$

Iteration (k)	w^k	$\nabla_w J = 2[w_1 \ w_2]$	J	$w^{k+1} = w^k - \alpha \nabla_w J$
0 <i>Init guess</i>	[3 4]	[6 8]	29	$[3 \ 4] - 2 * [6 \ 8] = [-9 \ -12]$
1	$[-9 \ 12]$	$[-18 \ 24]$	229	$[27 \ 36]$
2	$[27 \ 36]$	$[54 \ 72]$	2029	$[-81 \ 108]$

Gradient Descent example

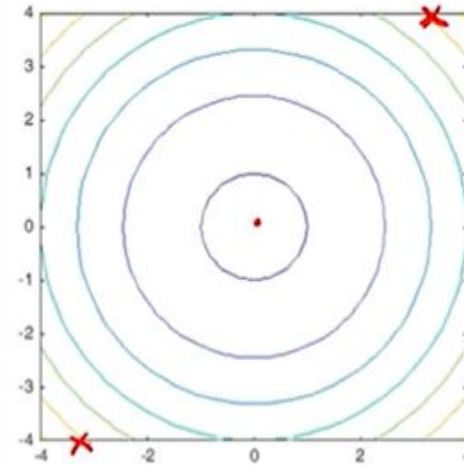


$$J(w) = w_1^2 + w_2^2 + 4$$

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$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

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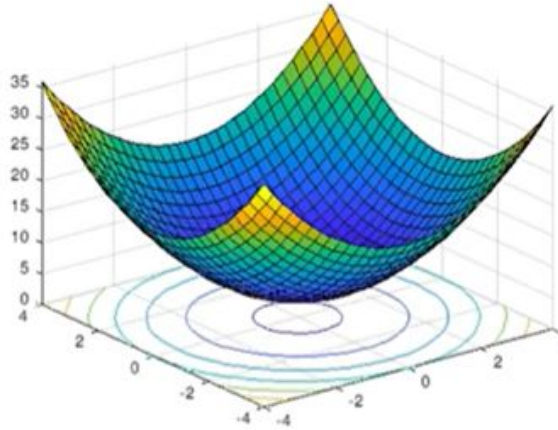


Oscillate

$$\mathbf{w}^0 = [3 \ 4]^T \quad \alpha = 1$$

Iteration (k)	$\mathbf{w}^k$	$\nabla_w J = 2[\mathbf{w}_1 \ \mathbf{w}_2]$	J	$\mathbf{w}^{k+1} = \mathbf{w}^k - \alpha \nabla_w J$
0	$[3 \ 4]$	$[6 \ 8]$	29	$[3 \ 4] - 1 * [6 \ 8] = [-3 \ -4]$
1	$-[3 \ 4]$	$-[6 \ 8]$	29	$[3 \ 4]$
2	$[3 \ 4]$	$[6 \ 8]$	29	$[-3 \ -4]$

Gradient Descent example

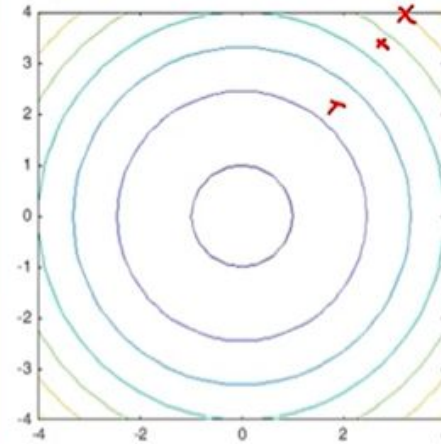


$$J(w) = w_1^2 + w_2^2 + 4$$

$$\nabla_w J(w) = \begin{bmatrix} 2w_1 \\ 2w_2 \end{bmatrix}$$

$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

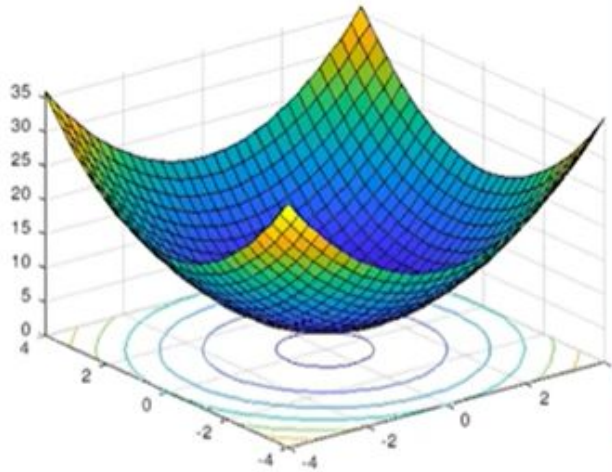
$$w_2^{k+1} = w_2^k - \alpha (2w_2^k)$$



$$w^0 = [3 \ 4]^T \quad \alpha = 0.1$$

Iteration (k)	w^k	$\nabla_w J = 2[w_1 \ w_2]$	J	$w^{k+1} = w^k - \alpha \nabla_w J$
0	[3 4]	[6 8]	29	$[3 \ 4] - 0.1 * [6 \ 8] = [2.4 \ 3.2]$
1	[2.4 3.2]	[4.8 6.4]	20	[1.92 2.56]
2	[1.92 2.56]	[3.84 5.12]	14.24	[1.536 2.048]
30	[0.0037 0.005]	...	4.0000	...

Gradient Descent example

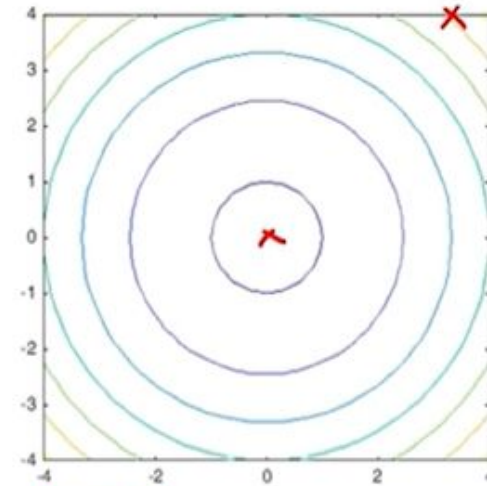


$$J(w) = w_1^2 + w_2^2 + 4$$

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$$w_1^{k+1} = w_1^k - \alpha (2w_1^k)$$

$$w_2^{k+1} = w_2^k - \alpha (2w_2^k)$$



$$w^0 = [3 \ 4]^T \quad \alpha = 0.5$$

$$w = w - \alpha \boxed{\nabla J}$$

Iteration (k)	w^k	$\nabla_w J = 2[w_1 \ w_2]$	J	$w^{k+1} = w^k - \alpha \nabla_w J$ $[0, 0] = [0, 0] - \alpha [0, 0]$
0	$[3 \ 4]$	$[6 \ 8]$	29	$[3 \ 4] - 0.5 * [6 \ 8] = [0 \ 0]$ ✓
1	$[0 \ 0]$	$[0 \ 0]$	4	$[0 \ 0]$
2	$[0 \ 0]$	$[0 \ 0]$	4	$[0 \ 0]$

Some lessons from the example

- It is possible for the gradient descent algorithm to
 - Diverge ($\alpha=2$)
 - Oscillate without diverging or converging ($\alpha=1$)
 - Converge slowly ($\alpha=0.1$)
 - Converge rapidly ($\alpha=0.01$)

